

Advanced Robot Control and Learning

Summer Semester 2026

Course Project

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Due: 07.09.2026

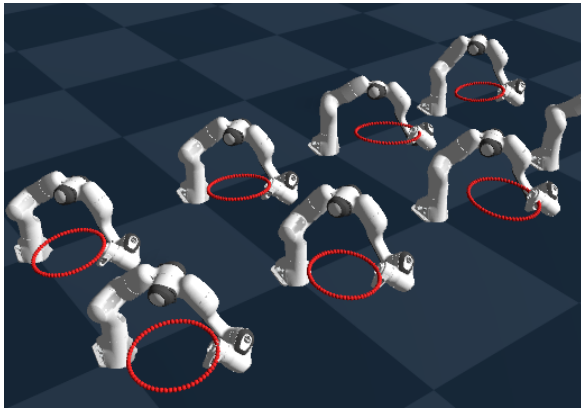


Figure 1: Example ARCL_WS folder structure

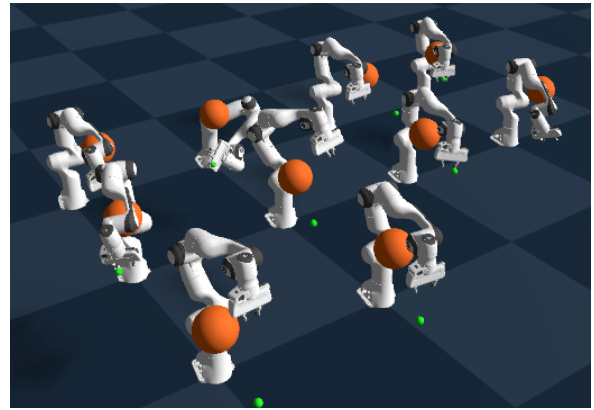


Figure 2: Control and RL example scripts

Submission Instructions:

- Submit a single PDF (**project report**) including the modelling assumptions, controller design, simulation results, and discussion.
 - The submission rules are specified in Section 1.
 - The Franka Emika Panda manipulator simulated in the **Genesis** framework is recommended. Other simulators or modelling environments, such as MuJoCo, PyBullet, Isaac Sim, or MATLAB/Simulink, may also be used.
 - Projects may be completed **individually or in teams of two**. Individual contributions must be clearly documented.
 - AI tools may be used as auxiliary support, but submitted solutions must remain technically consistent, justified, and understood by the student. Obviously auto-generated incorrect solutions may lead to substantial point deductions.
 - The project is graded on the German scale from **1.0 (best)** to **5.0 (fail)**.
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1 Project Overview

In this course project, you will implement and evaluate a control method for a robotic manipulator based on one selected research paper.

The **main objective** is to understand and reproduce the central method of the paper in simulation. A complete reproduction of all experiments in the original paper is not required.

1.1 Simulator

The **Franka Panda** manipulator serves as the main reference platform. Ready-to-use simulation templates based on the **Genesis** framework are provided during the course.

The choice of simulation environment is open. Suitable options include:

- Genesis,
- MATLAB/Simulink,
- MuJoCo,
- PyBullet,
- Isaac Sim,
- or another suitable simulation environment.

A planar or lower-dimensional robot model may also be used when it is sufficient to reproduce and evaluate the selected control method.

1.2 Topic Selection

Each project must:

- select one topic from Section 2;
- use the corresponding paper as the main technical reference;
- explain the main control formulation;
- implement the central method proposed in the paper;
- evaluate the method through simulation;
- discuss the obtained results and the students' own understanding of the method.

1.3 Presentation and Submission

- Projects may be completed **individually or in teams of two**.
- Each team will give a **15-minute presentation on (TBD)**:
 - 8 minutes presentation,
 - 7 minutes discussion and questions.
- A **2–4 page report** must be submitted using the IEEE double-column conference template¹ by **07.09.2026**.
- The complete codebase must be submitted as a **.zip** file no later than the presentation day (TBD).

¹<https://www.overleaf.com/latex/templates/ieee-conference-template/grfzhnncsfqn>

2 Project Topics

The project topics are fixed and predefined. Each project is based on one reference paper.

The main task is to reproduce the core controller and demonstrate its main properties in simulation.

2.1 Continuous Null-Space Projections for Hierarchical Control

Goal. Reproduce the torque-based hierarchical control method proposed in the paper, with continuous activation and deactivation of prioritized objectives.

Requirements.

- Use a redundant manipulator.
- Define at least two objectives with different priorities.
- Implement the torque-based hierarchical controller and the continuous null-space projection method proposed in the paper.
- Design a simulation in which one objective is activated or deactivated during the motion.
- Evaluate the task behavior and commanded joint torques.
- Explain why the continuous projection is used and discuss the behavior during task transitions.

Note. Possible additional investigations include comparison with a discontinuous task activation, analysis of torque continuity, or variation of the activation parameters. These are suggestions rather than mandatory requirements.

Reference paper.

- A. Dietrich, A. Albu-Schäffer, and G. Hirzinger, “On Continuous Null Space Projections for Torque-Based, Hierarchical, Multi-Objective Manipulation,” in *2012 IEEE International Conference on Robotics and Automation (ICRA)*, St. Paul, MN, USA, 2012, pp. 2978–2985, doi: [10.1109/ICRA.2012.6224571](https://doi.org/10.1109/ICRA.2012.6224571).

2.2 Multi-Priority Cartesian Impedance Control

Goal. Reproduce the quadratic-programming-based Cartesian impedance controller proposed in the paper for multiple tasks with different priorities.

Requirements.

- Use a redundant manipulator.
- Define at least two Cartesian impedance objectives with different priorities.
- Formulate and solve the hierarchical quadratic programs proposed in the paper.
- Include the joint-torque constraints in the controller.
- Design a simulation in which the objectives conflict or the torque constraints become active.
- Evaluate the task behavior, task priorities, and commanded joint torques.
- Explain how task priorities and torque constraints are handled by the optimization.

Note. Possible additional investigations include comparison with a classical null-space controller, removal of the torque constraints, or variation of the task weights and priority order. These are suggestions rather than mandatory requirements.

Reference paper.

- E. M. Hoffman, A. Laurenzi, L. Muratore, N. G. Tsagarakis, and D. G. Caldwell, “Multi-Priority Cartesian Impedance Control Based on Quadratic Programming Optimization,” in *2018 IEEE International Conference on Robotics and Automation (ICRA)*, Brisbane, QLD, Australia, 2018, pp. 309–315, doi: [10.1109/ICRA.2018.8462877](https://doi.org/10.1109/ICRA.2018.8462877).

2.3 Stability of Variable Impedance Control

Goal. Reproduce the variable impedance controller and investigate the stability conditions proposed in the paper.

Requirements.

- Implement a Cartesian or joint-space impedance controller.
- Introduce time-varying stiffness and damping parameters.
- Implement the stability condition proposed in the paper.
- Simulate at least one impedance profile that satisfies the condition and one profile that does not satisfy it.
- Evaluate the resulting motion and the variation of the impedance parameters.
- Explain why positive stiffness and damping alone are not sufficient to guarantee stability for time-varying impedance.

Note. Possible additional investigations include comparison with fixed impedance control, parameter sweeps, external disturbances, or a study of the conservativeness of the stability condition. These are suggestions rather than mandatory requirements.

Reference paper.

- K. Kronander and A. Billard, “Stability Considerations for Variable Impedance Control,” *IEEE Transactions on Robotics*, vol. 32, no. 5, pp. 1298–1305, Oct. 2016, doi: [10.1109/TRO.2016.2593492](https://doi.org/10.1109/TRO.2016.2593492).

3 Implementation Instructions

Example code templates based on the Genesis simulator are provided to help you get started. These templates are optional and may be modified or replaced.

3.1 Version Control

All project work should be version-controlled using Git.

```
git clone <repository_url>
git add .
git commit -m "Describe your change"
git push
```

3.2 Python Environment

The default project template requires Python 3.10 or newer.

Using uv (recommended).

```
uv venv
source .venv/bin/activate
uv pip install genesis-world torch numpy matplotlib
```

Using Conda (alternative).

```
conda create -n arcl_project python=3.10
conda activate arcl_project
pip install genesis-world torch numpy matplotlib
```

3.3 Genesis Resources

- Documentation: <https://genesis-world.readthedocs.io/>
- GitHub: <https://github.com/Genesis-Embodied-AI/Genesis>

3.4 Example Project Structure

```
ARCL_WS/
|-- Control_example/
|-- panda_cylinder.xml
```

The provided examples illustrate basic robot-control template and are intended only as starting points.

3.5 Robot and Tool Modeling

The file `panda_cylinder.xml` shows how the Franka Panda robot and a simple end-effector are specified using MJCF.

MJCF reference: <https://mujoco.readthedocs.io/en/stable/XMLreference.html>

```

<body name="attachment" pos="0 0 0.107" quat="0.3826834 0 0 0.9238795">
  <site name="attachment_site"/>
  <!-- === Square paddle === -->
  <body name="sensor" pos="0 0 0.05" quat="1 0 0 0">
    <geom
      name="sensor_geom"
      type="cylinder"
      size="0.01 0.055"
      rgba="0.2 0.6 0.9 1"
      density="700"
      friction="0.8 0.1 0.01"
      contype="1" conaffinity="1"
    />
  <body name="tool_tip" pos="0 0 0.055">
    <site name="tool_site" type="sphere" size="0.001" pos="0 0 0" rgba="1 0.3 0.3 1"/>

```

Figure 3: Example Genesis simulation screenshot using the Franka manipulator